1. How bfs is implemented

2. There r two large vessels filled with a litre of apple juice and grape juice. A tablespoonful of apple juice is poured into grape juice vessel.Then out of the mixture(apple and grape) one tablespoonful of the mixture is poured into apple juice vessel

Which of the following options are true

A apple juice is more in the grape juice compared to grape juice in apple juice

B grape juice is more in apple juice than compared to grape juice in apple juice

C Both r same

D Depends on the size of tablespoonful.

3. which of the following is not bst

We had to choose non bst from 4 figures.

4. construct a bst from empty tree

4.1 no of non leaf nodes in the bst

4.2 postorder and preorder traversal

5 no of nodes in a full binary tree if there r n nodes which r non leaf.

A 2^n -1

B 2^n + 1

C 2n +1 right

D 2n -1

6 AABCD,_________,ABCCCCD,ABCDDDDD fill in the blanks

7 24 bit instructions with 3 bit multiplier

A 192

B 256

C 320

D 384

8 in three address mode which of the following field is to be specified

Memory register

Processor register

________ accumulated register

Maths 9 2,7,14,23,___,47

Maths integration from -3 to 3 of |x +1|dx

A 5

B 10

C 15

D 20

11.given a no of disk accesses

Apply page replacement algorithm

1 farthest in future

2 first in first out

Give the sum of page faults and cold misses

12. Suppose there is a 2 digit no ending with d such that the product of the digit is same as the sum of the digits then the value of d is

A 5

B 9

C none of the above

D all the above

13. The return value of f(96870)

Int main()

{

m=0;

```
While(n>1)

{

m=10*m + n%10;

n=n/10;

}

Return m;

}
```

A 0

B

C

D none of the above

14. Random is a function which returns some value in the range (0,1). Where does the function converge if N is very large value.

```
Int N int count;

For (I=0;I<N;I++)

{

X=random (); y=random ();

If (x*x + y*y <= 1)

Count++;

}

Print("%f"float(count/N));
```

A pi/4

B pi

C (pi)^2

D does not converge

15. Two tricky comprehension

5 qs each

https://gateoverflow.in/127841/iiith-pgee-2017
https://gateoverflow.in/tag/iiith-pgee
PGEE 2016

Paper 1: Aptitude (45 questions, MCQ) 90 minutes, questions had multiple answers.

1. 2 english comprehensions each with 5 questions, no antonym and synonyms.
   level was good (not easy at all).

2. **3 questions on pnc, 3 questions of probability.**

3. 2 dbms questions on functional dependency, to find the key in the relation with the given functional dependencies.

4. many simple questions on basic programming or data structures concepts.

5. many questions on logical reasoning.

Paper 2: Technical (60 questions, MCQ) 90 minutes, questions had multiple answers

1. 5 questions of Operating system. All were easy, 1 from cpu scheduling, 1 from page fault service time,
   1 was deadlock charactestics, 2 from page replacement

2. 6 questions from DBMS.
   3 reations were given with the attributes and 5 questions on queries were asked, tuple calculus, relational algebra.
   1 question on BCNF and 3NF properties.

3. 12-13 C output questions were asked. (Basics should be clear, not very tricky ones.)
   Few questions were error recoganisation. 2-3 on pointers. few on arrays.

4. 4-5 questions from computer networks. (Easy)
   One question on transmission time and in other one propagation time.
   Protocol for electronic mail was asked (SMTP).
   No. of networks possible for class C in IPv4.

5. Algorithms
   1 question was which algorithm time complexity does not depend on size of the input.
   1 question was out of the given options, in which of the following insert, delete and findMin operation can be carried out in logn time.
   1 question was on optimal number of multiplications operations required to multiply 5 given

matrices.
1 was to solve recurrence relation
3-4 questions on BST construction and pre and post oreder traversal.

6. Mathematics
1 question was to find out the solution of the given equations.
few questions on relations
1 was possible numbers of non negative solution for x+y+z = 11.
2-3 questions on gcd.

7. Computer Organisation
1 was floating point conversion (to convert -1.5 in single precision floating point).
1 was to find out the square of binary number 1011 in hexadecimal.

Technical :
8 ques from DS
5 ques from C
4 ques from Digital
3 ques from OS
3 ques from discrete
1 ques from Probability
1 ques from COA
General Aptitude :
1)if there are 5 Men and 2 women find the number of ways in which these people can be made to sit around in a table(clockwise and anticlockwise seprately) with the two women not sitting with each other
options :
6!/2!
6!x2!240480

2) 2-3 problems on time,speed,distance(Trains and other)

3)Allegation problem

4)2 RC(Too long) having 6 ques each

5)2-3 paragraphs and options were given. Asked type: which of the following sentence weakens/strengthens the argument

6) LCM of two nos is 280 and their ratio is 7:8. Guess these numbers.

7)2-3 problems on time and work

9) problem on -difference between simple and compound interest.

10>Q.1 Sum of 1/4 + 1/9 + 1/16 .... is
a. less than 1b. greater than 1c. equal to 1d. (dont remember)*Akshay mane

11>. If 3 cards are drawn from a pack of 52 cards in how many ways it is possible that all three cards are of different types.
Options : a. 53C13b. 13*12*11c. 4*(13^3)and 1 more

13>1) find the least value of 3^x+3^(-x)a)2b)1c)0

4>10)if a finishes the work in 15 days and b finshes the work 10 days and if both work for 5 days and then b left the job then find the days for which a must work to finish the piece of work

15>trigonometry question: if shadow of a building when sun elevation is 45 degree is 20 m larger than that when sun's elevation is 60 degree..dont remember the option, very big options including square root 3..

16>if a=16*sq root2 and b=2*sq root2 , find log b(base a)? dont remember options ..

17> if a man can do a work in 15 days and a woman can do it in 20 days , after doing for 5 days woman leaves the work, in how many days man will complete the work.

18> in apti 2 ques n geometry relatd to circle was dere.

19)There are 6 locks and 6 keys .In how many attempts you can find out which lock matches which key

Subject Questions
1)Which of the following are true?
i)Stack can be used to create Queues.
ii)Queues can be used to create Stacks.
iii)Queues can be used to create stack but Stack cannot be used to create Queues.
iv)Stack can be used to create Queues but Queue cannot be used to create Stack
2)Sort the following in non decreasing order of their worst case search complexityBinary Search Tree,Hash Table, LinkedList, Array

3)You have a box containing 10 black and 10 blue socks.What is the minimum number of times you need to pull out so that you have a pair of the same color?
a)5
b)3c)8

d)11
4)You have an unbounded fan in AND gate,one unbounded fan in OR gate.How many AND gates will be required to represent a Boolean function of 5 variables.You can use as many NOT gates as possible.
5)A question on Dynamic Programming

6)A question on Ford Fulkerson Algorithm
To calculate maximum flow possible from source to destination
A graph was given.

7)Write -1 in 2's compliment in 16 bits and 8 bits

8)3-4 Guess 'C' code output

9) Thread/IPC related 4-5 sentences were given and and type was which sentences are true.A qsn from IPC is H/w support needed for message passing,pipes,shared memory...
10) What will be the output of the following code int count=0;

```
int increment(){return ++count;}
main()
{
int a[]={1,2,3,4,5};
a[count++]= increment();
printf("%d",a[count]);
}
```

a.1
b.2
c.3
d.4
e.5
11. Which of the following can be used to implement any digital circuita.2x1 MUX
b.4x1 MUX
c.8x1 MUX
d.16x1 MUX
e. None of these

12. The execution time of a program depends on

a. Clock frequencyb. No of Clocks per instructionc. No of memory accessd. No.of instructionse. Room temperature ....@akshay mane

13>Which of the following are truea. Message passing requires additional h/w
b. stdin/stdout are not implemented on all OS
c. Pipe on windows require h/w
d. Shared memory requires h/w

14>
```
for(i=0;i
{
sum= sum + A[i];
}
```
Which are true

a] sum exhibit temporal locality
b]array A exhibit Spatial locality
c] variable i exhibit temporal locality
d]
e]
*aisehi hi kuch to tha @akshay mane

15>What is the 2's complement representation of of -1 in 16 bit and 8 bit resp.
a] 0xFFFF, 0xFF
b]0x0000
c] 0x0001
d]
e]
*aisehi hi kuch to tha @akshay mane

16>let f be a function f(G,G')
which accepts two graphs G,G' and The SUBGRAPH ISOMORPHIC problem is "Wether Does the graph G contains a subgraph which is isomorphic to "

1. Problem is in P2. Problem is in NP3. Problem is in NP-Hard4. Problem is in exponential class
a] 1 only
b] 2,3 only
c]
d]
e]
*aisehi hi kuch to tha @akshay mane

17>An algorithm is said to be OBLIVIOUS if the Worst case complexity is equal to Best Case Complexity

i) Merge Sort ii) Bubble Sort iii) Quick Sort vi) Heap Sort

a] 1 only
b] 2 ONLY
c] 3 ONLY

d] 4 ONLY
e] 2,3 ONLY
*Dont remember exact abcde, @akshay mane

18>T(n) = 1+ Summation(k=1 to k=n-1) {T(k)+T(n-k)} n>1
= 1 n=1
a] ohm(n^2)
b] ohm(2^n)
c]
d]
e]
* all the ans were OHM(***) @akshay mane

19. Problem on Matrix Chain...Minimization problem
dimensions are given as <p0,p1,p2,p3,p4,p5> <20,75,25,95,110>

20>7)find the contrapostive of a sentence
sentence was like this .--> if i know the rules and i am not overconfident then i will won the game .. (not sure of this sentence)
21>hoow many maximum AND gates req to slove and eqution of varible 5.. use use OR gate
a>5
b>20
c>1024
d>*dont rember but somethingg of this sort.

There was a question that given a perimeter p,which one of these has least area?
1.Equilateral triangle
2.Right angled isosceles triangle
3.Circle
4.Square
5. Hexagon
So whats the correct answer ?
find side or radius in terms of perimeter p, then find area and compare. example: for circle 2.pi.R=P.... R=P/(2.pi)....area= pi.(R^2)=[pi/(4.pi*pi)].(P^2)....we can find this for all and compare the result.

basic pointer and array related questions (5-6 questions)... passing array as a parameter, using static variable in a recursive function call, assignment of individual items of an array to another array using * operator and indexes (trying to confuse), recursive function going for an infinite run etc..

process arrives at 0 2 6 times respectively with burst times of 10 20 30 respectively..find number of context switches in srtf scheduling ignoring context switches at beginning and end

one question of fork system call..it was also gate previous year question
if (fork() == 0)

{ a = a + 5; printf(“%d,%d\n”, a, &a); }
else { a = a –5; printf(“%d, %d\n”, a, &a); }
Let u, v be the values printed by the parent process, and x, y be the values printed by the child process. Which one of the following is TRUE?
(a) u = x + 10 and v = y
(b) u = x + 10 and v != y
(c) u + 10 = x and v = y
(d) u + 10 = x and v != y

time and work-2 probs, elementary probability- 3,4 questions, general arithmetic apti 5-6 questions
which app layer protocol uses non connection oriented network layer protocol. options were udp smtp, dns..
Consider the following relational schemes for a library database:
Book (Title, Author, Catalog_no, Publisher, Year, Price)
Collection (Title, Author, Catalog_no)

with in the following functional dependencies:

I. Title Author --> Catalog_no
II. Catalog_no --> Title Author Publisher Year
III. Publisher Title Year --> Price
Assume {Author, Title} is the key for both schemes. Which of the following statements is true?
(A) Both Book and Collection are in BCNF
(B) Both Book and Collection are in 3NF only
(C) Book is in 2NF and Collection is in 3NF
(D) Both Book and Collection are in 2NF only

Consider a B+-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?
(A) 1
(B) 2
(C) 3
(D) 4

Answer (B)

In a heap with n elements with the smallest element at the root, the 7th smallest element can be found in time (GATE CS 2003)
a) \theta(n log n)
b) \theta(n)
c) \theta(log n)
d) \theta(1)

Assume the following C variable declaration

int *A [10], B[10][10];
Of the following expressions
I A[2]
II A[2][3]
III B[1]
IV B[2][3]

which will not give compile-time errors if used as left hand sides of assignment statements in a C program (GATE CS 2003)?

a) I, II, and IV only
b) II, III, and IV only
c) II and IV only
d) IV only

Consider the pseudocode given below. The function DoSomething() takes as argument a pointer to the root of an arbitrary tree represented by the leftMostChild-rightSibling representation. Each node of the tree is of type treeNode.

```
typedef struct treeNode* treeptr;
struct treeNode {
treeptr leftMostChild, rightSibling;
};
int DoSomething (treeptr tree) {
int value = 0;
if (tree != NULL) {
if (tree->leftMostChild == NULL)
value = 1;
else
value = DoSomething(tree->leftMostChild);
value = value + DoSomething(tree->rightSibling);
}
return(value);
}
```

When the pointer to the root of a tree is passed as the argument to DoSomething, the value returned by the function corresponds to the
(A) number of internal nodes in the tree.
(B) height of the tree.
(C) number of nodes without a right sibling in the tree.
(D) number of leaf nodes in the tree.

The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)? (GATE CS 2004)
A)2
B) 3
C)4
D)6

https://gateoverflow.in/tag/iiith-pgee

The first paper had questions from probability theory (various probability distributions such as exponential, Bernoulli, geometric), coordinate geometry (equations of lines when they're rotated), linear algebra (mainly matrix algebra), computer programming (very basic questions involving predicting the output of a program), simple calculus (maxima and minima, slope), permutations and combinations, work and time, speed and distance, and a lot of other general aptitude questions. Most of the questions were straightforward and a lot of people were able to solve them. There were two reading comprehensions. Each of them had 4-5 questions that followed.